

TransFAD: Hybrid CNN–LSTM and Ensemble-Based Approach for Transmission Line Fault Detection and Classification

Abstract—In this paper, a hybrid TransFAD system, which abides by voltage and current measurements is introduced and uses these measurements as a basis to detect and classify transmission line faults. Protection of a power system involves fast and reliable fault detection. But in the real world, the noise and transient behavior as well as class imbalance can impair the operation of traditional approaches. Current methods utilize either handcrafted characteristics or individual learning models, allowing them to be inadequate to allow characterization of both sub-instantaneous electrical attributes and temporal fault dynamics. To overcome it, TransFAD integrates classical machine-learning-based classifiers with a CNN-LSTM-based deep learning model. Representation of system imbalance is done using engineered electrical features, and the CNN-LSTM learns spatial and temporal patterns directly on sequence of signal signals. A weighted ensemble strategy is used to combine the predictions of the individual models in order to enhance robustness and accuracy. The efficacy of TransFAD is shown with reference to a publicly accessible electrical faults detection dataset, in which it is assessed by different faults and non-fault conditions and contrasted with standalone models.

Index Terms—fault detection, classification, transmission line, ensemble, CNN-LSTM, SVM kernel, voltage, current, 1D CNN-BiLSTM, RMSE, MAE, R^2

I. INTRODUCTION

In this paper, we present a data driven framework for detecting and classifying faults in electrical transmission lines, a critical task for ensuring the reliability and stability of modern power systems. Transmission networks routinely experience disturbances caused by adverse weather, insulation breakdown, equipment aging, or external interferences. If these faults are not identified and isolated rapidly, they may propagate through the grid leading to equipment damage, large scale outages, and substantial economic loss. The increasing dependence on automation and intelligent grid infrastructure has motivated the transition from traditional threshold based fault detection methods to advanced computational approaches capable of handling the nonlinear and dynamic characteristics of power system signals.

In recent years, machine learning techniques have been widely studied for transmission line fault detection and classification. Early works employed shallow models such as k-nearest neighbors, decision trees, and support vector machines using voltage and current signals as features [2], [8], [11]. Samantaray *et al.* demonstrated that extreme learning machines provide fast and accurate fault detection with low computational complexity, making them suitable for real-time

protection [4]. Robust machine learning based schemes capable of handling noise and parameter variations were further explored in [3]. To improve classification performance, several studies combined signal processing with machine learning. Wavelet based feature extraction coupled with probabilistic neural networks and SVMs enhanced fault discrimination by capturing transient characteristics [10]. AI based fault classification and localization methods showed improved accuracy over conventional techniques [12], while decision tree based approaches highlighted the effectiveness of interpretable models for protection systems [8]. Subsequent research emphasized adaptability and generalization. Adaptive fault classification methods were proposed in [13], and supervised learning-based fault detection and localization for multi terminal systems were addressed in [14]. Single-ended directional fault identification schemes reduced reliance on synchronized measurements [9], while comparative studies highlighted the importance of feature selection and model optimization under diverse fault scenarios [6]. More recently, Ozupak presented a comparative analysis of machine learning models with random search based hyperparameter optimization [1], and broader reviews emphasized the need for data-driven protection strategies that generalize across varying operating conditions [7]. However, most existing approaches rely on a single learning paradigm, limiting their ability to jointly capture instantaneous electrical characteristics and latent temporal patterns. While these approaches offer interpretability and low computational cost, they rely primarily on static feature representations and struggle to model the temporal evolution of transient disturbances. Deep-learning models such as Convolutional Neural Networks (CNN) and Long Short Term Memory (LSTM) architectures have shown promise in capturing nonlinear and sequential behavior; however, deep models alone may fail to generalize under limited training data or significant class imbalance.

The proposed framework is evaluated using voltage and current datasets under various fault and non fault conditions. With the widespread deployment of measurement technologies such as phasor measurement units (PMUs) and high-resolution digital relays, large volumes of multi phase voltage and current data are readily available. However, these measurements often contain noise, transient fluctuations, and class imbalance, making conventional rule-based or handcrafted-feature methods insufficient for robust fault classification. Similar to the challenges encountered in other signal processing domains, including 3D digitization and reconstruction tasks, power

system fault detection requires the ability to extract meaningful structure from complex, high-dimensional raw data.

Towards this, we propose TransFAD, a hybrid framework that integrates traditional machine-learning classifiers such as Non Linear, Soft Margin Support Vector Machine with radial basis function (RBF) Kernel, with a spatial temporal deep learning architecture. In contrast to techniques which depend only on the presence of static products or only on deep representations, TransFAD integrates engineered electrical products and automatically learned temporal patterns to be able to classify steady state and transient fault signatures. Additionally, a combination of classical ML models and the CNN-LSTM network allows the use of the synergies of the strengths, which enhances the overall accuracy and noise resistance.

The paper has made the following contributions:

- We suggest TransFAD, a hybrid ML and DL system with problematic transmission lines fault detection and fault classification. We use state-of-the-art handcrafted electrical functionality to augment the principles of discrimination towards classical ML models.
- We design a CNN-LSTM architecture capable of learning spatial-temporal dependencies from multi phase voltage and current sequences.
- We introduce a weighted ensemble strategy that fuses predictions from Logistic Regression, SVM, Decision Tree classifiers, and the CNN-LSTM model to improve robustness and generalization.
- We conduct extensive experimentation using the Kaggle Electrical Fault Detection Dataset, including feature engineering, sequence modeling, normalization, and performance comparison across multiple metrics (Accuracy, F1-score, RMSE, MAE, and R^2).
- We demonstrate that TransFAD outperforms standalone ML and DL models, establishing it as a scalable and effective solution for real-time smart grid fault analytics.

In Section II, we describe the methodology, dataset and pre-processing pipeline. Section III outlines the proposed hybrid ML-DL methodology. Section IV presents the experimental setup and results, followed by conclusions in Section V

II. TRANSFAD: HYBRID CNN-LSTM ENSEMBLE FRAMEWORK

This work is proposing TransFAD, a hybrid transmission line fault detection and classification framework. As illustrated in Fig. 1, the framework is processing the same synchronized three-phase voltage and current measurements through two parallel inference branches: an engineered-feature machine-learning (EF-ML) branch and a CNN-LSTM spatial-temporal deep-learning branch. This parallel processing is being adopted because transmission line faults are simultaneously exhibiting instantaneous electrical deviations and short-duration temporal dynamics. The complementary information extracted from both branches is being combined using weighted probability fusion to improve classification robustness.

A. Input Signal Representation

Each observation is being represented by the synchronized three-phase voltage and current vector

$$\mathbf{x} = [V_a, V_b, V_c, I_a, I_b, I_c] \in \mathbb{R}^6, \quad (1)$$

where V_a , V_b , and V_c are denoting the instantaneous phase voltages of phases a , b , and c , respectively, and I_a , I_b , and I_c are denoting the corresponding phase currents measured at the relay location. This representation is being used because transmission line faults are primarily manifesting as voltage depression, current surge, and phase imbalance, all of which are directly observable in these six signals. The associated class label is being denoted by

$$y \in \{1, 2, \dots, K\}, \quad (2)$$

where K is denoting the total number of operating conditions, including normal and faulty states.

B. Engineered-Feature Machine-Learning (EF-ML) Branch

The EF-ML branch, shown in Fig. 3, is explicitly extracting physically interpretable electrical features from the raw measurements and is mapping them to fault probabilities using classical machine-learning classifiers. This branch is being included to provide explainable and computationally efficient fault evidence that complements deep temporal modeling.

1) *Engineered Electrical Feature Builder*: Inter-phase voltage asymmetry is being captured using line-to-line voltage differences:

$$L_{Vab} = V_a - V_b, \quad L_{Vbc} = V_b - V_c, \quad L_{Vac} = V_a - V_c, \quad (3)$$

where L_{Vab} , L_{Vbc} , and L_{Vac} are quantifying phase-to-phase voltage imbalance. These features are being used because asymmetrical faults are producing unequal voltage drops across phases.

Overall voltage depression and current surge behavior are being summarized using three-phase magnitudes:

$$V_m = \sqrt{V_a^2 + V_b^2 + V_c^2}, \quad I_m = \sqrt{I_a^2 + I_b^2 + I_c^2}, \quad (4)$$

where V_m is denoting the resultant voltage magnitude and I_m is denoting the resultant current magnitude. These quantities are being used to compactly represent overall fault severity.

Phase imbalance is being quantified using dispersion across phases:

$$V_{imb} = \text{std}(V_a, V_b, V_c), \quad I_{imb} = \text{std}(I_a, I_b, I_c), \quad (5)$$

where $\text{std}(\cdot)$ is denoting the standard deviation operator. These features are being included because balanced operating conditions are producing low dispersion, whereas faults introduce significant imbalance.

All engineered features are being concatenated to form

$$\phi(\mathbf{x}) = [\mathbf{V}, \mathbf{I}, \Delta\mathbf{V}, V_m, I_m, I_{imb}, V_{imb}] \in \mathbb{R}^{13}, \quad (6)$$

where $\mathbf{V} = [V_a, V_b, V_c]$, $\mathbf{I} = [I_a, I_b, I_c]$, and $\Delta\mathbf{V} = [L_{Vab}, L_{Vbc}, L_{Vac}]$.

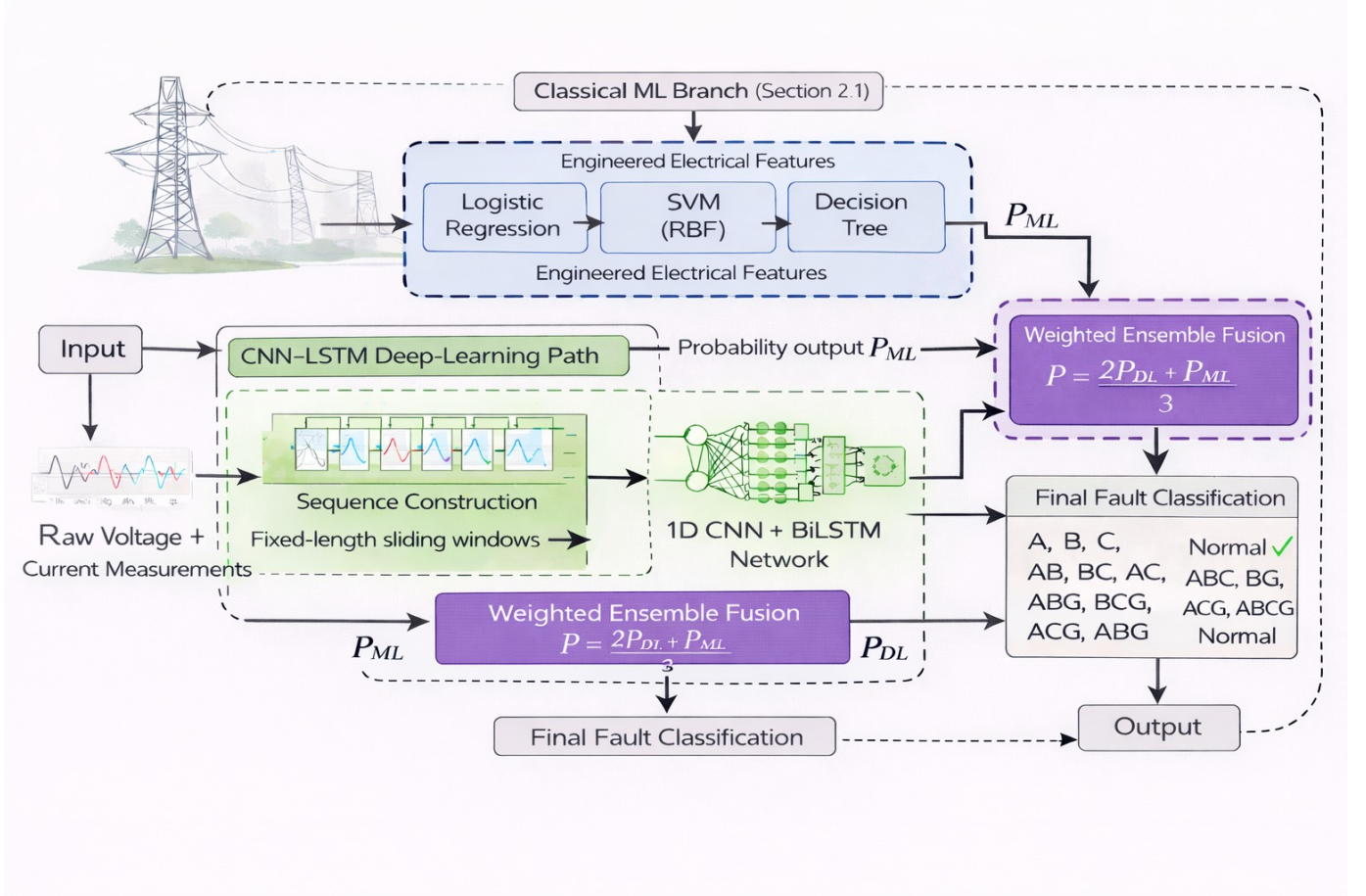


Fig. 1. Illustration of proposed framework TransFAD: Hybrid CNN LSTM

2) *Feature Standardization and Classical Classification:* Each engineered feature is being standardized as

$$\mathbf{z} = \frac{\phi(\mathbf{x}) - \boldsymbol{\mu}}{\boldsymbol{\sigma} + \epsilon}, \quad (7)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are denoting feature-wise mean and standard deviation estimated from training data, and ϵ is a small constant preventing numerical instability. Standardization is being applied to ensure equal feature contribution during optimization.

The standardized feature vector $\mathbf{z}$ is being processed by Logistic Regression with L2 regularization, RBF-kernel Support Vector Machine, and Decision Tree classifiers. These models are being selected to jointly capture linear trends, nonlinear boundaries, and rule-based decisions. The EF-ML branch is producing the class-posterior probability vector

$$\hat{\mathbf{P}}_{ML} \in \mathbb{R}^K. \quad (8)$$

C. CNN-LSTM Spatial-Temporal Branch

The CNN-LSTM branch, shown in Fig. 2, is directly learning spatial-temporal fault signatures from short sequences of raw measurements. This branch is being included because many fault characteristics are transient and evolve over time.

1) *Sequence Construction and Windowing:* A fixed-length input sequence is being constructed as

$$\mathbf{X} \in \mathbb{R}^{T \times 6}, \quad T = 50, \quad (9)$$

where T is denoting the number of consecutive time samples and each row is containing the six-channel measurement vector. The window length $T = 50$ is being chosen to capture transient fault evolution while maintaining low inference latency.

2) *Synthetic Temporal Augmentation and Normalization:* To improve generalization, additive noise is being injected during training:

$$\mathbf{X}'(t, i) = \mathbf{X}(t, i) + \epsilon_{t,i}, \quad \epsilon_{t,i} \sim \mathcal{N}(0, 0.01), \quad (10)$$

where $\epsilon_{t,i}$ is denoting Gaussian noise emulating sensor uncertainty. Channel-wise normalization is then being applied to stabilize deep-network training.

3) *1D CNN Feature Extraction:* Local temporal features are being extracted using 1D convolution:

$$Z_k(t) = \sum_{i=1}^6 \sum_{\tau=0}^{m-1} W_{k,i,\tau} X(t - \tau, i) + b_k, \quad (11)$$

where $W_{k,i,\tau}$ are learnable convolution weights, b_k is a bias term, and m is the kernel size. Stacked Conv1D layers with

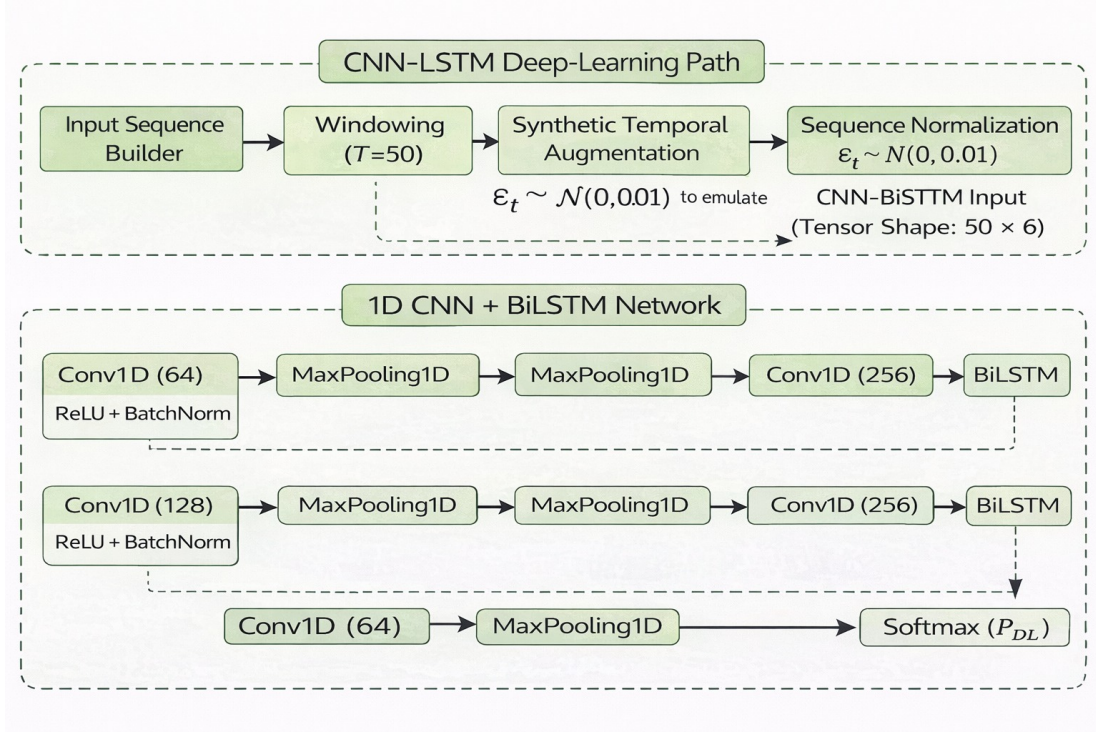


Fig. 2. CNN-LSTM deep-learning branch of TransFAD showing sequence windowing ($T = 50$), stacked 1D CNN-BiLSTM modeling, and softmax-based fault probability output P_{DL} .

increasing filter depth are being used to learn hierarchical fault representations.

4) *BiLSTM Temporal Modeling and Deep Output*: Bidirectional LSTM is being applied as

$$\mathbf{H}_t = [\vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t], \quad (12)$$

to capture forward and backward temporal dependencies. Mean pooling is being applied to obtain a fixed-length embedding

$$\mathbf{h} = \frac{1}{T'} \sum_{t=1}^{T'} \mathbf{H}_t, \quad (13)$$

which is being classified using softmax to produce

$$\hat{\mathbf{P}}_{DL} \in \mathbb{R}^K. \quad (14)$$

D. Weighted Ensemble Decision Fusion

As illustrated in Fig. 1, the posterior probabilities from both branches are being fused as

$$\hat{\mathbf{P}} = \frac{w_{DL} \hat{\mathbf{P}}_{DL} + w_{ML} \hat{\mathbf{P}}_{ML}}{w_{DL} + w_{ML}}, \quad (15)$$

where $w_{DL} = 2$ is emphasizing temporal modeling capability and $w_{ML} = 1$ is retaining interpretable engineered-feature evidence. The final predicted class is being obtained using

$$\hat{y} = \arg \max_c \hat{\mathbf{P}}(c). \quad (16)$$

III. RESULTS AND DISCUSSIONS

In this section, we provide a brief description on the dataset considered for experimentation, the experimental setup of the proposed methodology, and demonstrate the results of proposed methodology in comparison with state-of-the-art methods.

A. Dataset

We employ a publicly available *Electrical Fault Detection* dataset sourced from Kaggle to evaluate the proposed TransFAD framework. The dataset consists of synchronized three-phase voltage and current measurements recorded under normal and faulty operating conditions. Each sample is represented by six electrical parameters,

$$\{V_a, V_b, V_c, I_a, I_b, I_c\},$$

along with a binary label indicating fault or non-fault state.

The dataset reflects practical grid measurement challenges such as noise, overlapping feature distributions, and class imbalance, making it suitable for validating fault detection performance under realistic conditions.

1) *Data Preparation and Feature Construction*: All records with missing or inconsistent values are removed, and an 80–20 train–test split is performed using stratified sampling to preserve class proportions. Standard normalization is applied to stabilize training.

To enhance separability for classical classifiers, additional electrical features are derived from the raw measurements, including line-to-line voltages, voltage and current magnitudes,

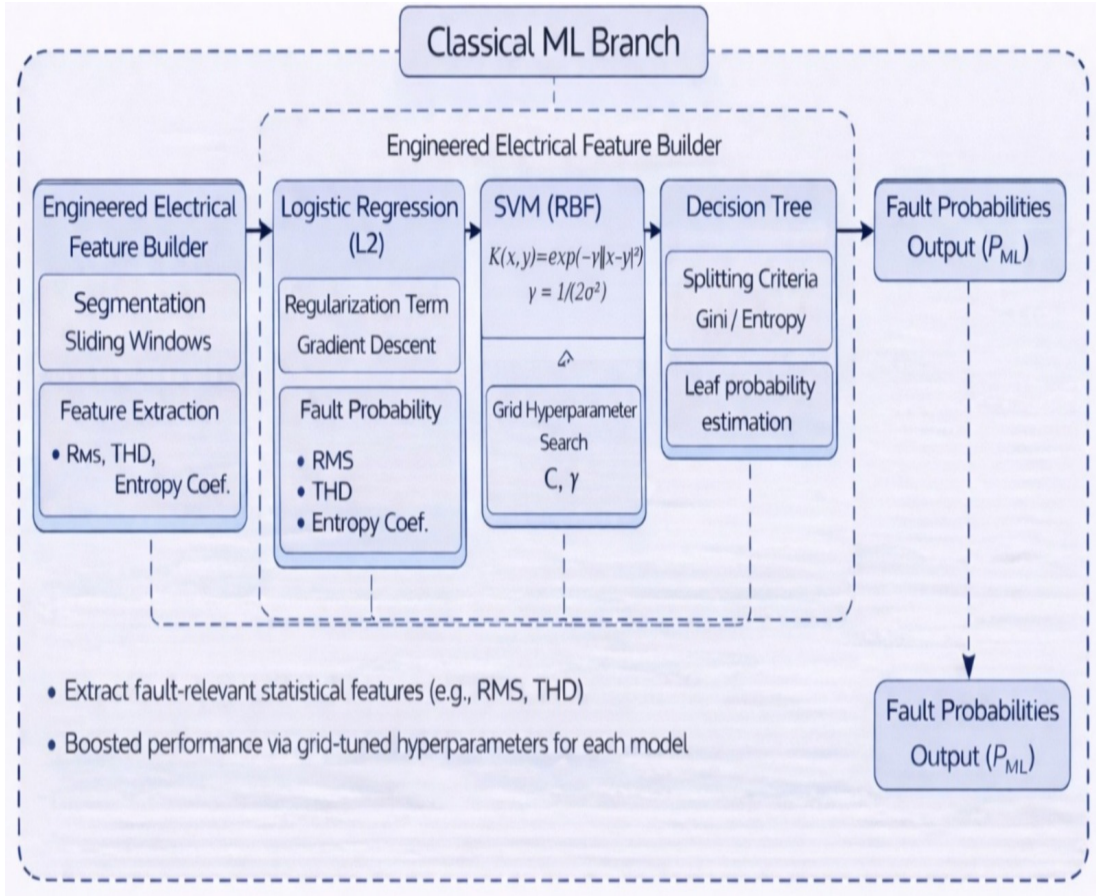


Fig. 3. Classical Machine Learning branch of TransFAD illustrating engineered electrical feature extraction, nonlinear classifiers, and probabilistic fault output P_{ML} .

and phase imbalance indicators. This expands the original six-dimensional input into a 12-dimensional feature vector.

2) *Sequence Generation*: As the dataset does not explicitly provide temporal sequences, fixed-length sequences of 50 timesteps are synthetically generated to enable temporal modeling. Small stochastic perturbations are introduced to simulate natural grid variations, and the resulting sequences are independently normalized before being fed to the CNN-LSTM network.

B. Experimental Setup

In this section, we describe the experimental configuration used to evaluate the proposed TransFAD framework, including training strategies for classical machine learning models, deep learning architecture, and the ensemble fusion mechanism.

1) *Classical Machine Learning Models*: We train three classical machine learning classifiers using the engineered 12-dimensional electrical feature set:

- **Logistic Regression**: trained with maximum iterations set to 1000 and class balancing enabled to mitigate label imbalance.
- **Support Vector Machine (RBF)**: configured with $C = 10$ and $\gamma = \text{scale}$, with probability estimation enabled and class balancing applied.

- **Decision Tree**: trained with class weighting to avoid bias toward the majority class.

All models are trained using scikit-learn with default settings unless specified otherwise and produce posterior class probabilities for downstream fusion.

2) *CNN-LSTM Training*: To capture temporal dependencies, a CNN-LSTM network is trained using synthetically generated sequences of length 50 timesteps derived from the raw electrical measurements. The network is trained using the Adam optimizer with a learning rate of 1×10^{-3} and cross-entropy loss. Training is performed with a batch size of 64 and a bidirectional LSTM hidden size of 128 units. Dropout regularization is applied at the input and fully connected layers to prevent overfitting. The network is trained for 5 epochs, sufficient to demonstrate convergence on the generated sequences.

3) *Ensemble Configuration*: Predictions from the classical machine learning models and the CNN-LSTM network are combined using a weighted probability fusion strategy. Given machine learning probabilities $\hat{P}_{ML}$ and deep learning probabilities $\hat{P}_{DL}$, the final fused probability is computed by assigning higher weight to the deep learning output to emphasize temporal modeling capability.

4) *Evaluation Metrics*: The performance of the proposed framework is evaluated using standard classification metrics, including Accuracy and F1-score. In addition, regression-based error metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2) are reported to assess prediction consistency and numerical error characteristics.

5) *Implementation Details*: All experiments are conducted using Python-based machine learning libraries, with scikit-learn employed for classical models and PyTorch for CNN-LSTM implementation. GPU acceleration is utilized where available to ensure efficient training and reproducibility.

C. Results

In this section, we demonstrate the results of the proposed TransFAD framework on electrical fault classification using the experimental setup described in the previous section. Due to the absence of standardized benchmarks for hybrid ML-DL fault diagnosis models operating on engineered electrical features with temporal learning, we present a comparative analysis across multiple ensemble configurations and evaluate their performance using standard classification and error-based metrics.

1) *Overall Performance Comparison*: Table I summarizes the quantitative performance of all three ensemble models.

The Decision Tree + CNN-LSTM ensemble achieves the highest performance across all evaluation metrics. The integration of CNN-LSTM enhances fault detection by capturing temporal variations, while classical ML models contribute structured decision boundaries that complement deep-learning representations.

2) *Accuracy and F1-Score Comparison*: Figure 4 shows a comparison of Accuracy and Macro F1-score across all ensemble models. All three models exceed 98% performance, with the Decision Tree ensemble achieving the best overall scores.

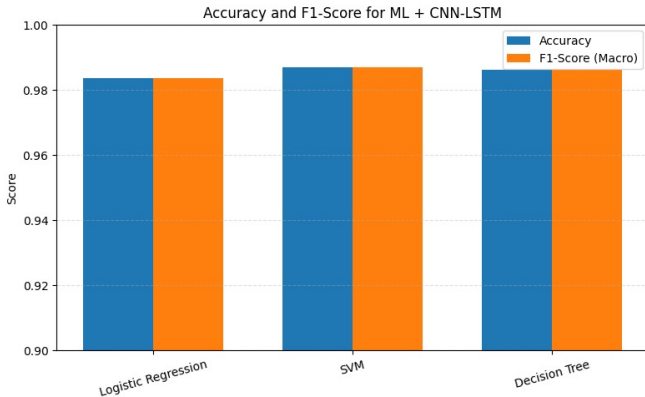


Fig. 4. Accuracy and F1-score comparison for Logistic Regression, SVM, and Decision Tree ensembles with CNN-LSTM.

3) *Confusion Matrix Analysis*: The confusion matrices provide insight into class-wise prediction behavior and misclassification patterns across ensemble models.

TABLE I
METHOD-WISE PERFORMANCE COMPARISON USING METRICS

Method	Study	Acc. (%)	RMSE	MAE
SVM	Özüpak (2025)	74.19	0.877	0.388
	Proposed	98.92	0.104	0.011
Logistic Regression	Özüpak (2025)	97.01	0.691	0.119
	Proposed	98.75	0.108	0.012
Decision Tree	Özüpak (2025)	99.87	0.017	0.005
	Proposed	98.96	0.102	0.010

TABLE II
PERFORMANCE COMPARISON OF ML + CNN-LSTM ENSEMBLE MODELS

Model	Acc	F1	RMSE	MAE	R^2
LR + CNN-LSTM(ours)	0.9875	0.9874	0.1118	0.0125	0.9497
SVM + CNN-LSTM(ours)	0.9892	0.9891	0.1041	0.0108	0.9564
DT + CNN-LSTM(ours)	0.9896	0.9895	0.1020	0.0104	0.9581

4) *Logistic Regression + CNN-LSTM*: Figure 5 illustrates the confusion matrix for the Logistic Regression + CNN-LSTM ensemble. The model exhibits low misclassification, with a small number of fault instances misclassified as non-fault.

Confusion Matrix: Logistic Regression + CNN-LSTM

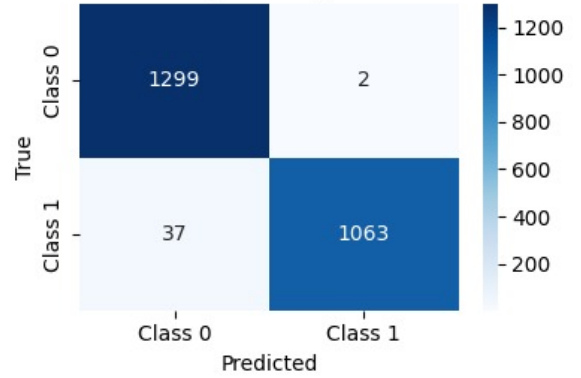


Fig. 5. Confusion matrix for Logistic Regression + CNN-LSTM ensemble.

5) *SVM + CNN-LSTM*: Figure 6 presents the confusion matrix for the SVM + CNN-LSTM ensemble. Compared to Logistic Regression, the SVM-based model demonstrates reduced false positives and false negatives.

6) *Decision Tree + CNN-LSTM*: Figure 7 shows the confusion matrix for the Decision Tree + CNN-LSTM ensemble. The model achieves the lowest misclassification rate among all evaluated ensembles, indicating balanced and stable classification behavior.

7) *Error Metrics Evaluation*: Although fault detection is primarily a classification task, regression-style error metrics provide additional insight into prediction stability and numerical consistency. As shown in Table I, RMSE and MAE decrease progressively from Logistic Regression to SVM to Decision Tree ensembles, while the R^2 score increases correspondingly.

Figure 8 illustrates the comparison of RMSE, MAE, and R^2 values across all ML + CNN-LSTM ensemble models.

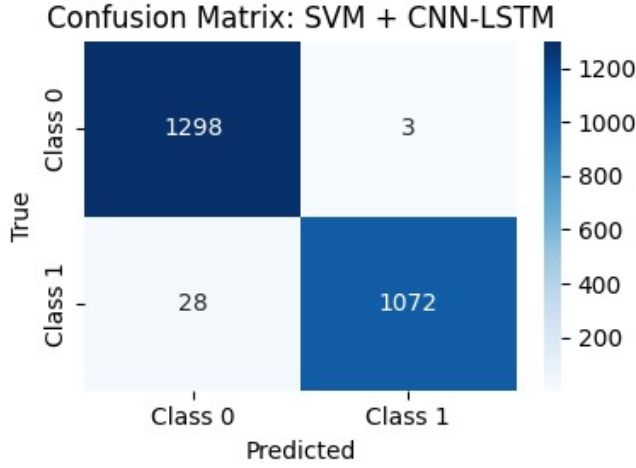


Fig. 6. Confusion matrix for SVM + CNN-LSTM ensemble.

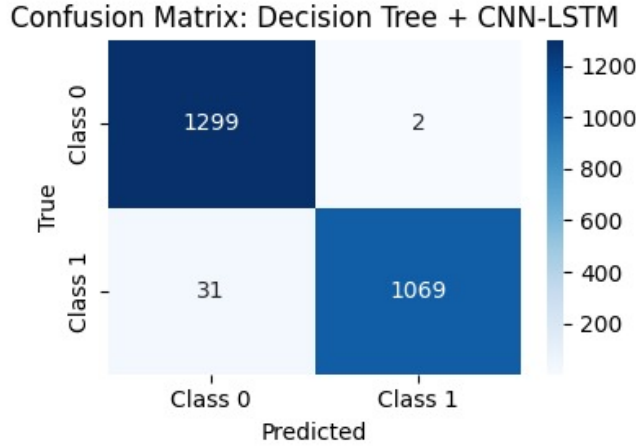


Fig. 7. Confusion matrix for Decision Tree + CNN-LSTM ensemble.

D. Discussions

The experimental findings reveal that the proposed TransFAD framework is working at promoting the combination of the classical machine-learning and deep-learning paradigms in the direction of electrical fault diagnosis. The CNN-LSTM element uses spatial-temporal interactions mandatory to identify a transient fault with the aid of classical ML classifiers that make decisions based on complex deterministic features with a view to increasing the robustness of the model.

The enhanced results of the Decision Tree + CNN-LSTM ensemble indicate that there is a complementary action between the partitioning using rules and profound time framework modeling. The reliability and scalability of TransFAD, ensured by the high accuracy and the low error levels, reveal that the diagnostic model is credible and suitable to use in the real-time fault analysis in a smart grid.

The experiment results obtained indicate successful integration of the classical machine learning with deep learning paradigms using the proposed TransFAD grounded on diag-

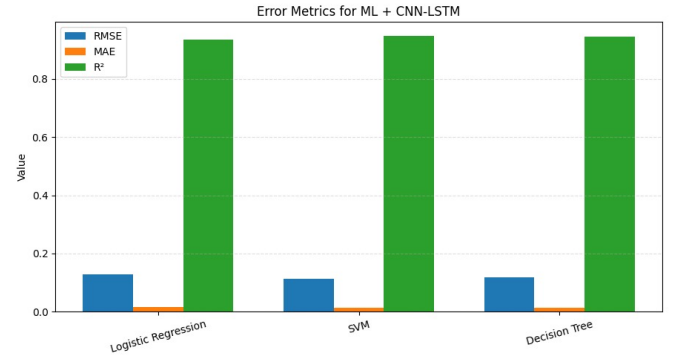


Fig. 8. Comparison of RMSE, MAE, and R^2 metrics across ML + CNN-LSTM ensemble models.

nosing electric faults. CNN-LSTM component helps to capture the desired spatial-temporal variations existing in electrical signals and the classical classifiers to have ordered, feature-driven decision boundaries that are more robust.

The higher performance of the Decision Tree + CNN-LSTM ensemble is an indication that no nodes work contrary to other groups in that respects of rule based feature partitioning versus deep temporal model. The reliability (monotony of precision, low errors) of TransFAD is reflected in the fact that the given model is a suitable and effective means of intelligent criticism to occur in a smart grid environment.

IV. CONCLUSION

This paper has designed a hybrid TransFAD system using a three phase voltage and current measurement to detect and classify transmit line faults. The framework suggested has covered the major issues linked to conventional fault detection methods such as sensitivity to noise, time-dependent behaviour and class skew. The existing approaches that were based on handcrafted characteristics or individual learning models were found to be insufficient in the measurement of both transient electrical properties and temporal fault behavior. To address these shortcomings, TransFAD proposed to use a CNN-LSTM based deep learning framework on top of classical machine learning classifiers. The system imbalance was modeled via engineered electrical features, and the CNN-LSTM model directly learned the patterns of space and time as a result of sequences of voltage and current signal. The predictions of individual models were amalgamated through application of weighted ensemble strategy, which gave rise to ability to detect as well as increased robustness. The efficacy of the suggested TransFAD framework was shown through a publicly area-outlined electrical fault detection information at fault and non-fault working conditions. The comparative analysis with the standalone models showed that the hybrid ensemble method performed at the best resulting in the capability of the hybrid ensemble method to give quality and dependable transmission line fault detection. These findings affirmed that a combination of classical machine learning and deep temporal learning is a

potential way forward to intelligent protection schemes in the contemporary power systems.

ACKNOWLEDGMENT

The authors have recognized the use of publicly available electrical fault detection datasets in order to conduct experiments. The language refinement and code development support were done with the help of AI-assisted tools whereas all the model design, experimentation and interpretation of results were done by the authors.

REFERENCES

- [1] Y. Özüpak, "Machine learning-based fault detection in transmission lines: A comparative study with random search optimization," *Bull. Polish Acad. Sci., Tech. Sci.*, vol. 73, 2025, doi: 10.24425/bpasts.2025.153229.
- [2] A. H. Ali, A. M. Abdulsada, and M. T. Yaseen, "Detection of power transmission line faults based on voltages and currents values using k-nearest neighbors," *IEEE Access*, vol. 9, pp. 135217–135228, 2021, doi: 10.1109/ACCESS.2021.3115412.
- [3] A. A. Salam, M. S. Rahman, and A. K. M. M. Islam, "Robust fault detection and classification in power transmission lines using machine learning techniques," *Electr. Power Syst. Res.*, vol. 189, p. 106663, 2020, doi: 10.1016/j.epsr.2020.106663.
- [4] S. R. Samantaray, P. K. Dash, and G. Panda, "Fast and accurate fault detection and classification in transmission lines using extreme learning machine," *IET Gener., Transm. Distrib.*, vol. 3, no. 5, pp. 431–441, 2009, doi: 10.1049/iet-gtd.2008.0472.
- [5] Y. Özüpak, "Transmission line fault detection and classification using artificial intelligence techniques," in *Proc. Int. Conf. Power System Transients (IPST)*, 2019, pp. 1–6.
- [6] R. K. Patel and A. Panahi, "Transmission line fault detection using machine learning approaches," *IEEE Trans. Power Del.*, vol. 35, no. 6, pp. 2914–2923, Dec. 2020, doi: 10.1109/TPWRD.2020.2998735.
- [7] J. L. Sarmiento, J. C. Delfino, and E. R. Arboleda, "Machine learning advances in transmission line fault detection: A literature review," *Preprints*, 2024, doi: 10.20944/preprints202405.0265.v1.
- [8] S. R. Samantaray, "Decision tree-based fault classification in transmission lines using current signals," *IEEE Trans. Power Del.*, vol. 25, no. 3, pp. 1547–1555, Jul. 2010, doi: 10.1109/TPWRD.2009.2035530.
- [9] A. Yadav and A. Swetapadma, "A single-ended directional fault section identification scheme for transmission lines using machine learning," *IEEE Trans. Power Del.*, vol. 34, no. 2, pp. 579–588, Apr. 2019, doi: 10.1109/TPWRD.2018.2876485.
- [10] M. Biswal, P. K. Dash, and S. Mishra, "Transmission line fault detection and classification using wavelet transform and probabilistic neural network," *IEEE Trans. Power Del.*, vol. 27, no. 1, pp. 467–475, Jan. 2012, doi: 10.1109/TPWRD.2011.2171002.
- [11] P. Ray, D. P. Mishra, and S. R. Samantaray, "Fault classification in transmission lines using support vector machines," *IEEE Trans. Power Del.*, vol. 26, no. 4, pp. 2626–2635, Oct. 2011, doi: 10.1109/TPWRD.2011.2157673.
- [12] S. A. Elkalashy, T. Kawady, and N. I. Elkalashy, "Artificial intelligence-based fault classification and location in transmission lines," *IET Gener., Transm. Distrib.*, vol. 5, no. 5, pp. 525–533, May 2011, doi: 10.1049/iet-gtd.2010.0289.
- [13] M. J. B. Reddy and D. K. Mohanta, "Adaptive fault classification using machine learning techniques for transmission line protection," *IEEE Trans. Smart Grid*, vol. 7, no. 4, pp. 1982–1991, Jul. 2016, doi: 10.1109/TSG.2015.2501421.
- [14] H. Livani and C. Y. Evrenosoglu, "A fault classification and localization method for three-terminal circuits using machine learning," *IEEE Trans. Power Del.*, vol. 28, no. 4, pp. 2282–2290, Oct. 2013, doi: 10.1109/TPWRD.2013.2260173.
- [15] S. Mishra, C. N. Bhende, and B. K. Panigrahi, "Detection and classification of power system disturbances using SVM-based approach," *IEEE Trans. Power Del.*, vol. 25, no. 3, pp. 1719–1727, Jul. 2010, doi: 10.1109/TPWRD.2009.2037227.